

23ADC04 DEEP LEARNING

Mark Split-Up (Total: 100 Marks)

The project is evaluated across 10 phases covering the entire lifecycle from ideation to GitHub upload. The marking rubric is designed to reward both technical depth and process discipline.

Phase / Activity	Marks	Deliverable / Assessment Criteria
1. Project Proposal & Title Approval	5	1-page individual abstract with objective, dataset, methodology, expected output
2. Literature Survey	10	Minimum 5 papers reviewed individually; gap analysis and justification table
3. Dataset Collection & Preprocessing	10	Dataset description, EDA notebook, preprocessing pipeline in code, done independently
4. Model Architecture Design	10	Block diagram, layer-by-layer explanation, justification of architecture
5. Model Implementation (Code)	15	Full working code authored solely by the student, modular structure, requirements.txt
6. Training & Validation Results	15	Training curves, confusion matrix, accuracy/loss plots, evaluation metrics
7. GitHub Repository Setup & Upload	10	Personal GitHub repo with proper README, folder structure, .gitignore, commit history (min 5 commits)
8. Project Report (Documentation)	10	Full IEEE-format individual report covering all modules, results & conclusion
9. Project Demonstration & Viva	10	Individual live demo, Q&A on code, architecture, and deep learning concepts
10. Innovation & Real-world Impact	5	Novelty of approach, comparison with baseline, deployment readiness
TOTAL	100	Full project lifecycle from proposal to GitHub upload

Standard Operating Procedure (SOP)

Step	Activity	Timeline	Details & Deliverables
Step 1	Individual Title Selection	Week 1	Each student individually browses the 65 titles and selects ONE unique title matching their interest. Confirm with faculty. No two students in the same class may select the same title — first come, first allotted.
Step 2	Create Personal GitHub	Week 1	Create a public GitHub repo under your own account named: DL_<ProjectTitle>_<RollNo>. This is your individual repository — do not add

	Repository		classmates as collaborators. Create default branches: main and dev.
Step 3	Project Proposal Submission	<i>Week 2</i>	Create a proposal document (1 page) individually: Title, Roll No, Objective, Dataset source, Architecture overview, Expected outcome. Submit to faculty and upload as PROPOSAL.md on your GitHub.
Step 4	Literature Survey	<i>Weeks 3-4</i>	Individually review minimum 5 IEEE/Springer papers. Fill the Literature Survey Table (paper title, year, method, result, gap). Upload as LITERATURE_SURVEY.md with proper citations.
Step 5	Dataset Collection & EDA	<i>Weeks 4-5</i>	Independently collect/download dataset (Kaggle, UCI, HuggingFace, etc.). Perform EDA in Jupyter Notebook. Plot class distribution, visualize samples, check missing values. Commit as notebooks/EDA.ipynb.
Step 6	Preprocessing Pipeline	<i>Week 5-6</i>	Write your own preprocessing scripts: normalization, augmentation, tokenization, etc. Save as src/preprocess.py. Document all steps in README under 'Data Preprocessing' section.
Step 7	Model Design & Architecture	<i>Week 6-7</i>	Individually design and draw the model architecture (use draw.io or matplotlib). Save diagram as docs/architecture.png. Implement model in src/model.py covering all 3 course modules.
Step 8	Model Training & Evaluation	<i>Weeks 7-9</i>	Train model on your own, log metrics (accuracy, F1, AUC). Plot training/validation curves. Save trained model as models/best_model.h5. Add evaluation notebook as notebooks/Evaluation.ipynb.
Step 9	GitHub Repository Structuring	<i>Week 9</i>	Ensure repo follows the standard folder structure (see below). Write a comprehensive README.md in your own words. Add .gitignore for Python. Minimum 5 meaningful commits authored by you, with proper messages.
Step 10	Final Report Writing	<i>Weeks 9-10</i>	Write your own IEEE-format report: Abstract, Introduction, Related Work, Methodology, Results, Conclusion. Save as docs/FINAL_REPORT.pdf. Upload to GitHub under docs/ folder.
Step 11	Demo Video / Screenshot	<i>Week 10</i>	Record a 3-5 min individual screen demo OR take screenshots of running output. Upload as assets/demo.gif or assets/screenshots/. Embed in README.
Step 12	Final GitHub Upload & Submission	<i>Week 10</i>	Ensure all files are committed under your own account. Tag the final release: git tag v1.0.0. Submit your personal GitHub URL and roll number to faculty portal. Repo must be PUBLIC.

README.md Must Include

- Project title, your name & roll number
- Problem statement in 2-3 sentences
- Architecture diagram (embed docs/architecture.png)
- Dataset source link and description
- How to run: step-by-step commands
- Results table (accuracy, F1-score, AUC, etc.)
- Module mapping: how M1, M2, M3 concepts are used
- References (minimum 5 papers)

Git Commit Guidelines

Minimum 5 commits required. Each commit message must describe what was done:

1. feat: Add preprocessing pipeline for image normalization
2. feat: Implement CNN-LSTM hybrid model architecture
3. train: Complete model training — accuracy 94.3%
4. docs: Add literature survey and architecture diagram
5. final: Upload complete project report and evaluation results

Required Metrics by Problem Type

Project topics span many different deep learning problem types, so a single fixed set of metrics cannot be applied to every project. Students must identify their problem category below and report the corresponding metrics in the Training & Validation Results phase and in the README results table. If a project spans multiple categories, report metrics for all applicable categories.

Problem Type	Required Metrics
Image / Audio / Text Classification (binary or multi-class)	Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC-AUC (for binary tasks)
Object Detection (e.g., YOLO-based projects)	mAP (mean Average Precision), IoU, Precision-Recall curve, Inference speed (FPS)
Image Segmentation	IoU, Dice Coefficient, Pixel Accuracy
Regression / Time Series Forecasting	MAE, RMSE, MAPE, R^2 Score
NLP — Text Generation / Summarization	BLEU, ROUGE, Perplexity
NLP — Sentiment / Text	Accuracy, F1-score, Precision, Recall, Confusion

Classification	Matrix
Anomaly Detection (Autoencoder-based)	Precision, Recall, F1-score, AUC-ROC, Reconstruction Error
Generative Models (GAN-based)	FID (Fréchet Inception Distance), Inception Score, qualitative visual assessment
Recommendation Systems	Precision@K, Recall@K, NDCG, RMSE (for rating prediction)
Reinforcement Learning (e.g., drone navigation, DQN)	Cumulative Reward, Episode Length, Success Rate, Convergence plots
Speech / Audio Recognition	Word Error Rate (WER), Accuracy
Biometric / Face Recognition	Accuracy, False Accept Rate (FAR), False Reject Rate (FRR), Equal Error Rate (EER)

Dataset & Tool Resources

7.1 Public Dataset Platforms

- Kaggle — www.kaggle.com/datasets (Healthcare, Finance, NLP, Vision, Agriculture)
- UCI Machine Learning Repository — archive.ics.uci.edu
- HuggingFace Datasets — huggingface.co/datasets (NLP tasks)
- Google Dataset Search — datasetsearch.research.google.com
- PhysioNet — physionet.org (ECG, EHR medical datasets)
- SIIM / Grand Challenge — grand-challenge.org (Medical image datasets)
- COCO Dataset — cocodataset.org (Object detection, segmentation)
- OpenImages — storage.googleapis.com/openimages (Large-scale vision)

7.2 Recommended Tech Stack

- Python 3.10+, TensorFlow 2.x / PyTorch 2.x
- Keras, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
- OpenCV (image processing), NLTK / spaCy / Transformers (NLP)
- Git & GitHub Desktop (version control)
- Google Colab or Kaggle Notebooks (free GPU)
- Weights & Biases (wandb.ai) for experiment tracking

7.3 Reference Textbooks

6. Goodfellow, Bengio & Courville — Deep Learning (MIT Press, 2016) — deeplearningbook.org
7. Chollet, F. — Deep Learning with Python (Manning, 2nd Ed., 2021)
8. Geron, A. — Hands-On ML with Scikit-Learn, Keras, TensorFlow (O'Reilly, 3rd Ed., 2022)
9. Zhang et al. — Dive Into Deep Learning (Cambridge, 2023) — d2l.ai

Important Notes for Students

1. The GitHub repository must be PUBLIC, owned by the student's account, at the time of submission.
2. Code plagiarism between students (copying another student's repository) will result in ZERO marks for both students involved.
3. Minimum dataset size: 1000 samples.
4. Model accuracy alone does NOT guarantee full marks - process, documentation, and code quality matter equally.
5. The GitHub commit history is reviewed - bulk commits at the deadline will be penalized; commits must show genuine individual progress over time.

Submission Checklist

Use this checklist to confirm every requirement is complete before the final submission deadline.

- Title selected from the approved list and confirmed with faculty (no duplicate titles within the class)
- Personal GitHub repository created, named DL_<ProjectTitle>_<RollNo>, with main and dev branches
- PROPOSAL.md uploaded with title, roll number, objective, dataset source, architecture overview, and expected outcome
- LITERATURE_SURVEY.md uploaded with minimum 5 IEEE/Springer papers, gap analysis, and proper citations
- EDA notebook (notebooks/EDA.ipynb) committed, including class distribution, sample visualization, and missing value checks
- Preprocessing pipeline (src/preprocess.py) committed and documented in the README under 'Data Preprocessing'
- Architecture diagram (docs/architecture.png) and model implementation (src/model.py) committed
- Model trained and evaluated, with the metrics required for the project's problem type (see Required Metrics by Problem Type) reported
- Trained model saved (models/best_model.h5) and evaluation notebook (notebooks/Evaluation.ipynb) committed
- README.md includes all required sections: title/name/roll number, problem statement, architecture diagram, dataset link, run instructions, results table, module mapping (M1/M2/M3), and references
- .gitignore added for Python, and repository follows the standard folder structure
- Minimum 5 meaningful commits made under the student's own account, with descriptive messages

- Final IEEE-format report (docs/FINAL_REPORT.pdf) completed and uploaded under docs/
- Demo video or screenshots (assets/demo.gif or assets/screenshots/) recorded and embedded in the README
- Final release tagged as v1.0.0
- Repository visibility set to PUBLIC
- Dataset contains a minimum of 1000 samples
- No code or repository sharing with classmates; all work is individually authored
- GitHub repository URL and roll number submitted to the faculty portal before the deadline